

Prof. Dr. Boas Pucker

Pathway Databases

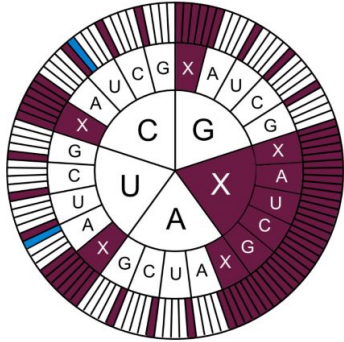
Availability of slides

- All materials are freely available (CC BY) - after the lectures:
 - eCampus: WBIO-A-08
 - GitHub: <https://github.com/bpucker/teaching>
- Questions: Feel free to ask at any time
- Feedback, comments, or questions: [pucker\[a\]uni-bonn.de](mailto:pucker[a]uni-bonn.de)

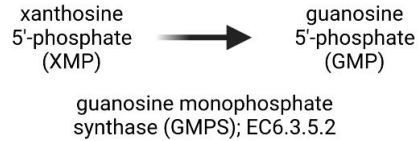


My figures and content can be re-used in accordance with CC BY 4.0, but this might not apply to all images/logos. Some figure were constructed using bioRender.com.

Why do we need pathway databases?



Expanding the genetic code with an additional DNA base (X)



<https://www.genome.jp/pathway/ec00230+6.3.5.2>

Identification of candidate enzymes in a **database**



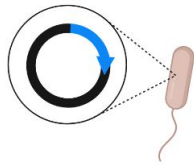
Croton tiglium is naturally producing a precursor of this unnatural DNA base



RNA extraction, sequencing, and construction of a transcriptome assembly



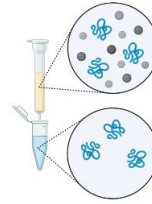
Cloning of GMPS coding sequence into a plasmid



Transformation into *Escherichia coli*



Heterologous expression of *GMPS*

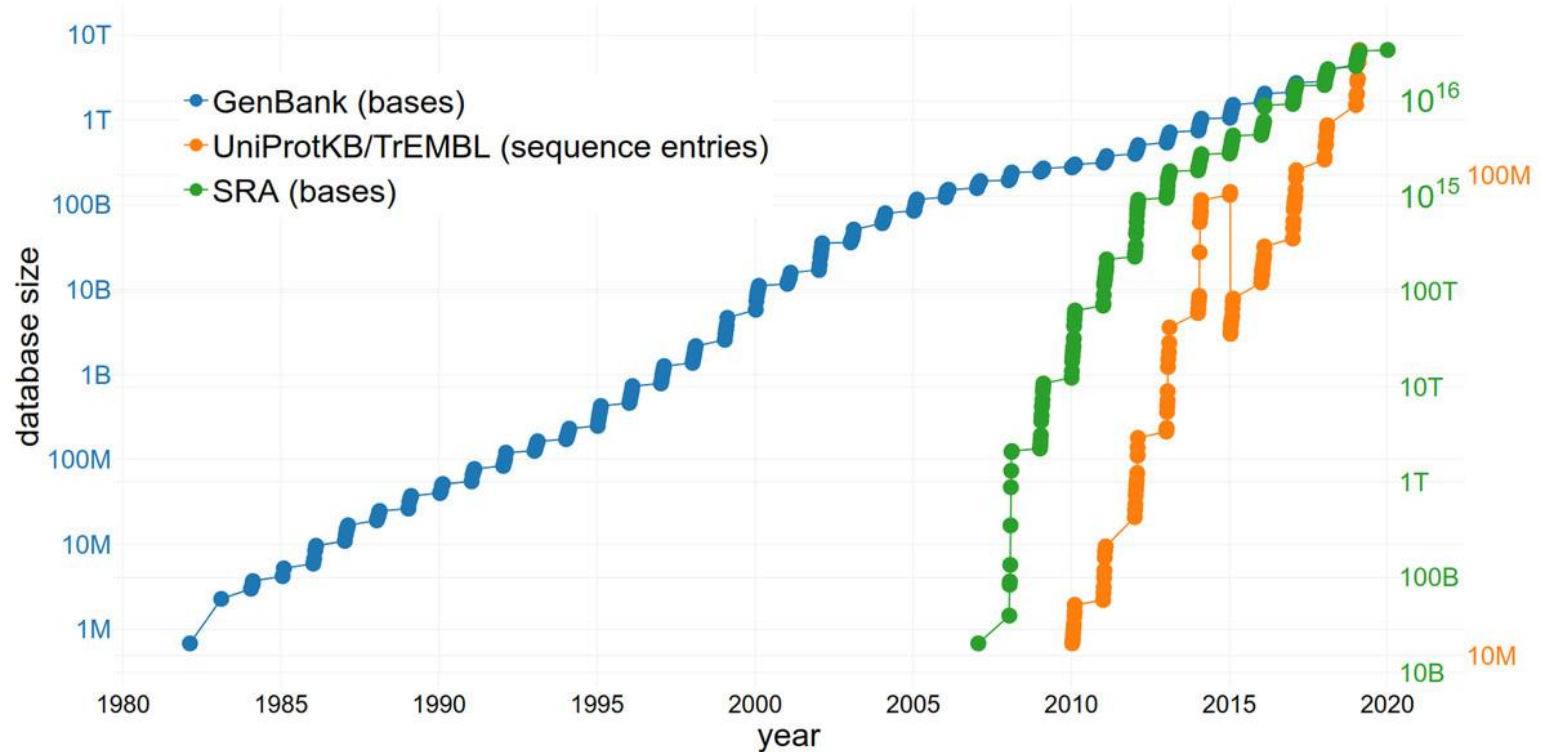


Protein purification



Enzyme assay

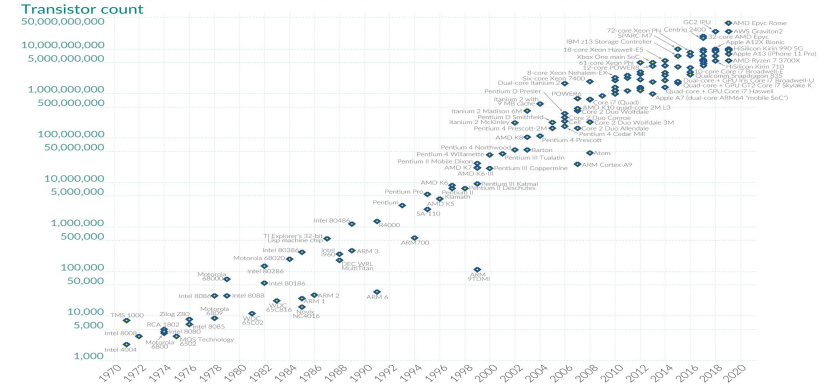
Exponential growth of databases



Generation of large data sets

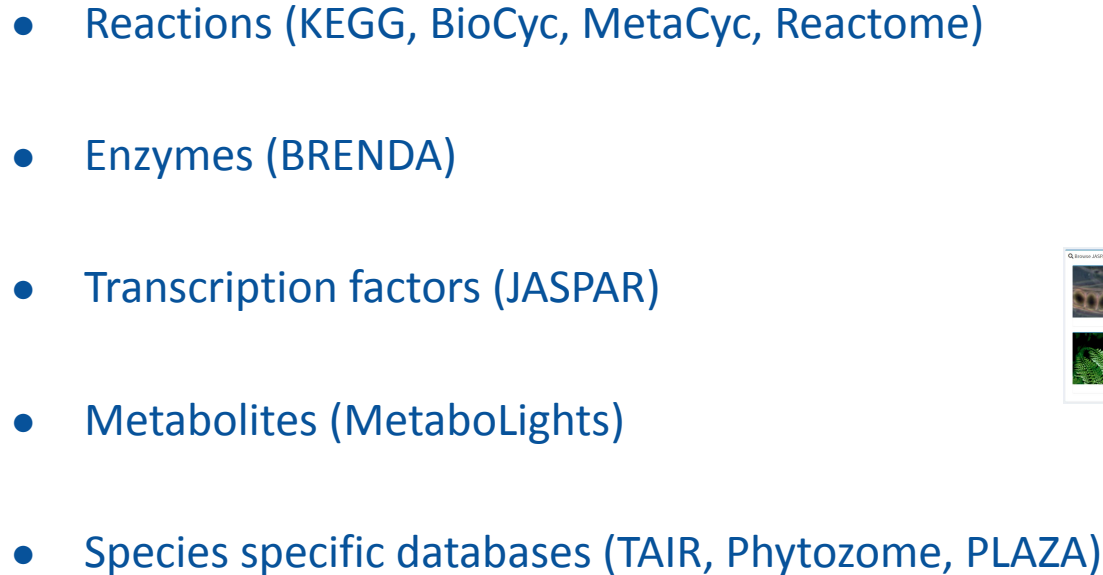
- Sequencing technologies are outpacing Moore's law
 - Moore's law = number of transistors on microchips doubles every two years
- Sequencing is becoming widely available (democratization of sequencing)
- Proteomics and metabolomics are developing rapidly

Moore's Law: The number of transistors on microchips doubles every two years Our World in Data
 Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)
 OurWorldInData.org – Research and data to make progress against the world's largest problems. Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.

https://commons.wikimedia.org/wiki/File:Moore%27s_Law_Transistor_Count_1970-2020.png



How to access data?

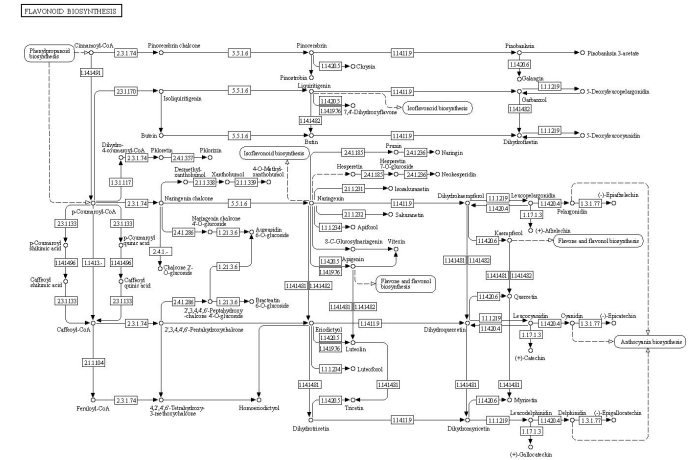
- Open Data = Initiative to make research data publicly available
- FAIR data = Findable Accessible Interoperable Reuseable data
 - Findable = registered in a central database
 - Accessible = available through a repository (not just from authors)
 - Interoperable = data format allows automatic processing
 - Reusable = sufficient metadata to use data sets in other studies
- ORCID is used as a universal login solution across many platforms
 - ORCID = Open Researcher and Contributor ID
 - SSO = Single Sign-On (user stays logged in across websites)



<https://www.go-fair.org/>

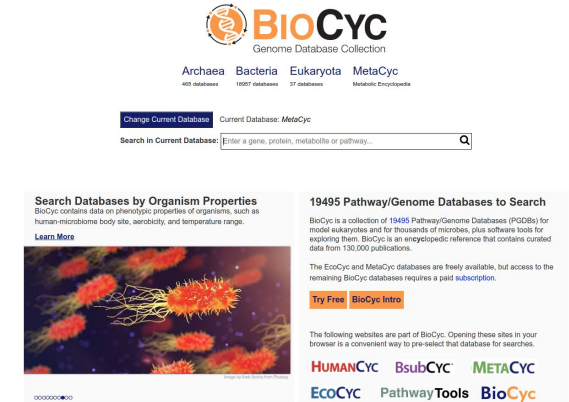


<https://orcid.org/>



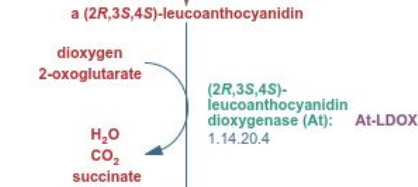
map00941 Flavonoid biosynthesis

- Integration of databases and tools for omics
- Curation is involved to increase the quality
- Strong focus on microorganism:
 - EcoCyc = **E**scherichia **coli** **E**ncyclopedia
 - BsubCyc = **B**acillus **sub**tilis **E**ncyclopedia
 - YeastCyc = **Y**east **E**ncyclopedia
- BUT:
 - Search for metabolites
 - Search for pathways
 - Search for reactions



- Integrates genomic data with functional annotation
- Visualization of pathway databases
- Shows intermediates and enzymes of biosynthesis pathways
- MetaFlux: flux-balance analysis

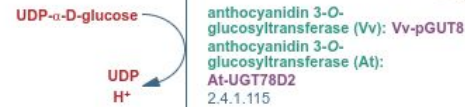
leucopelargonidin and leucocyanidin biosynthesis



a (4S)-2,3-dehydroflavan-3,4-diol



an anthocyanidin with a 3-hydroxy group



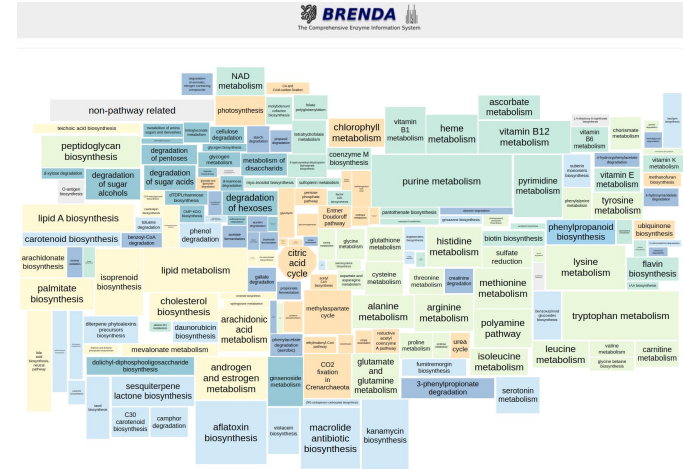
an anthocyanidin-3-O-β-D-glucoside

cyanidin dimalonylglucoside biosynthesis
anthocyanidin sophoroside metabolism
shisonin biosynthesis

anthocyanidin modification (*Arabidopsis*)
acylated cyanidin galactoside biosynthesis
rose anthocyanin biosynthesis II
(via cyanidin 3-O-β-D-glucoside)
rose anthocyanin biosynthesis I
(via cyanidin 5-O-β-D-glucoside)

Enzyme Commission (EC) numbers

- Four digit system to classify enzymes:
 - General type of reaction
 - Subclass
 - Sub-subclass
 - Serial number (within sub-subclass)
- General types:
 - EC1 = Oxidoreductases
 - EC2 = Transferases
 - EC3 = Hydrolases
 - EC4 = Lyases
 - EC5 = Isomerases
 - EC6 = Ligases



BRENDA - example (2)

- Specific substrates and products are listed
- K_M and V_{max} values are included
- References point to publications and other databases

top print hide 4 entries Go to Substrate/Product Search

SUBSTRATE A Y	PRODUCT A Y	REACTION DIAGRAM	ORGANISM A Y	UNIPROT A Y	COMMENTARY (Substrate) A Y	LITERATURE (Substrate) A Y	COMMENTARY (Product) A Y	LITERATURE (Product) A Y	Reversibility (irreversible) A Y	Not specified A Y
dihydrokaempferol + NADPH + H ⁺	leucopelargonidin + NADP ⁺	⊖	Arabidopsis thaliana	P51102	-	702000	-	-	?	
dihydroeriodictin + NADPH + H ⁺	eriodictin + NADP ⁺	⊖	Arabidopsis thaliana	P51102	-	702000	-	-	?	
dihydroquercetin + NADPH + H ⁺	quercetin + NADP ⁺	⊖	Arabidopsis thaliana	P51102	-	702000	-	-	?	
additional information	?	-	Arabidopsis thaliana	P51102	does not catalyze naringenin	702000	-	-	?	

top print hide Go to Cofactor Search

COFACTOR A Y	ORGANISM A Y	UNIPROT A Y	COMMENTARY A Y	LITERATURE A Y	IMAGE
NADPH	Arabidopsis thaliana	P51102	-	702000	⊖

top print hide 4 entries Go to Specific Activity Search

SPECIFIC ACTIVITY [μmol/min/mg] A Y	ORGANISM A Y	UNIPROT A Y	COMMENTARY A Y	LITERATURE A Y
0.0002	Arabidopsis thaliana	P51102	with dihydrokaempferol as substrate	702000
0.006	Arabidopsis thaliana	P51102	with eriodictin as substrate	702000
0.001	Arabidopsis thaliana	P51102	with dihydroeriodictin as substrate	702000
0.0013	Arabidopsis thaliana	P51102	with dihydroquercetin as substrate	702000

top print hide Go to Organism Search

ORGANISM A Y	COMMENTARY A Y	LITERATURE A Y	UNIPROT A Y	SEQUENCE DB A Y	SOURCE A Y
Arabidopsis thaliana	-	702000	P51102	UniProt	BRENDA

top print hide Go to General Information Search

GENERAL INFORMATION A Y	ORGANISM A Y	UNIPROT A Y	COMMENTARY A Y	LITERATURE A Y
physiological function	Arabidopsis thaliana	P51102	DFR plays a key role in determining intensity and pigment coloration because its specificity and activities dictate the type and amount of the colorless leucoanthocyanidins	702000

top print hide Go to AA Sequence and Transmembrane Helices Search Go to Localization Prediction

UNIPROT A Y	ENTRY NAME A Y	ORGANISM A Y	NO. OF AA A Y	NO. OF TRANS. HELICES A Y	MOLECULAR WEIGHT [kDa] A Y	SOURCE A Y	SEQUENCE A Y	LOCALIZATION PREDICTION A Y
P51102	DFR_ARATH	Arabidopsis thaliana	382	0	42775	Swiss-Prot	Show Sequence	Other Location (Reliability: 5)

top print hide Go to Cloned (Commentary) Search

CLONED (Commentary) A Y	ORGANISM A Y	LITERATURE A Y
into pTetHis2-TOPO and heterologously expressed in Escherichia coli TOP10F ⁺ strain. DFR cDNA cloned into pRSF-FHT and inserted into Escherichia coli BL21Star to create E-color strain	Arabidopsis thaliana	P51102 702000

top print hide References Search

REF. A Y	AUTHORS A Y	TITLE A Y	JOURNAL A Y	VOL. A Y	PAGES A Y	YEAR A Y	ORGANISM (UNIPROT) A Y	PUBLISHED A Y	SOURCE A Y
702000	Leonard, E.; Yan, Y.; Chen, J.; Ma, M.; U. Martens, S.; Koffas, M.	Characterization of dihydroflavonol 4-reductases for recombinant plant pigment biosynthesis applications	Biochemical Biotechnology	26	243-251	2008	Fragaria x ananassa (Q22517), Ipomoea nil (Q04607), Arabidopsis thaliana (P51102), Rosa hybrid cultivar (Q41158), Lilium sp. (Q8UAG7), Anthurium andraeanum (Q84L22)	-	BRENDA

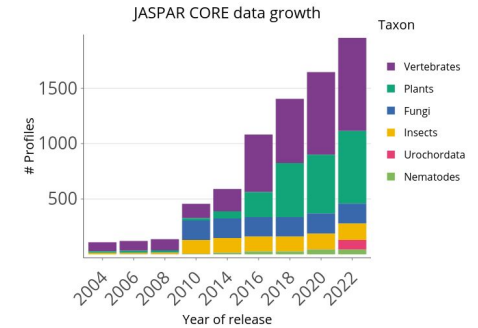
← Select items on the left to see more content.

top print hide 11 entries

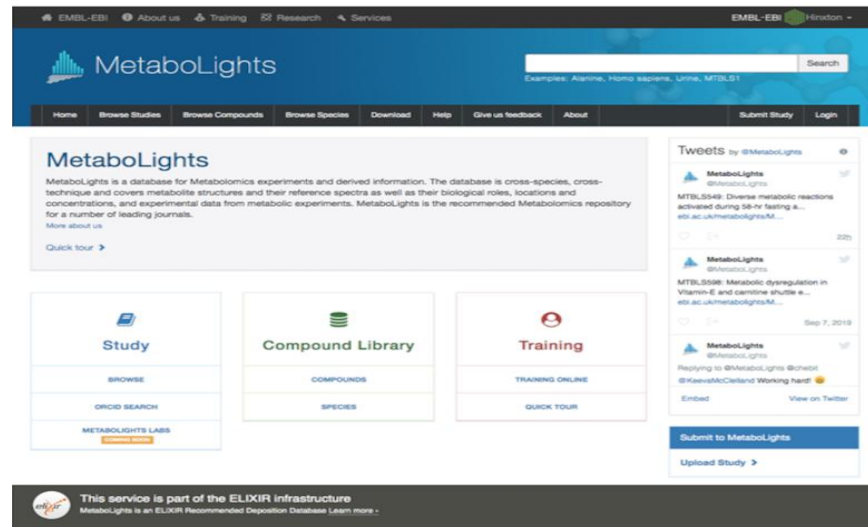
EXTERNAL LINKS (specific for EC-Number 1.1.1.219)
Expasy/TrEMBL (official portal for IUBMB Enzyme Nomenclature)
Enzyme Nomenclature Database
KEGG
MetaCyc
SABIO-RK
NCBI: PubMed, Protein, Nucleotide, Structure, Gene, OMIM
Enzyme Nomenclature (alternative site)
UniProt
PDB
PROSITE Database of protein families and domains
InterPro (database of protein families, domains and functional sites)

JASPAR: THE transcription factor database

- Contains motifs of transcription factor binding sites
- Additional details about transcription factors
- Most comprehensive database
- Regularly updated

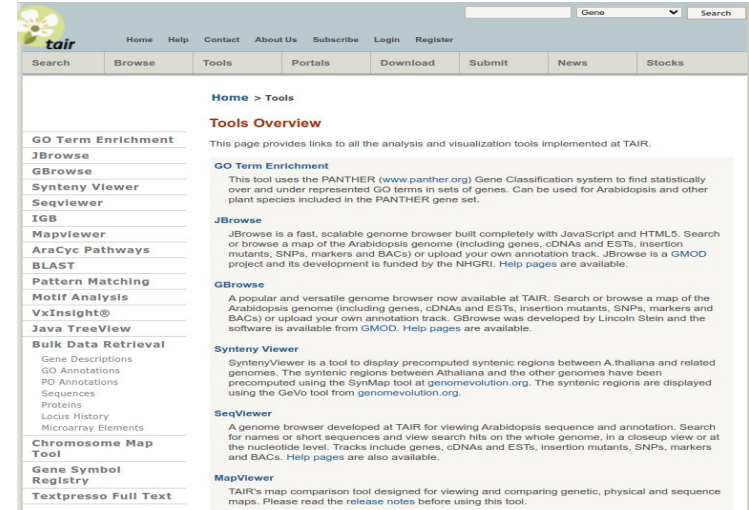


- Database of metabolomics experiments
- Hosted at European Bioinformatics Institute (UK)
- Collaboration with journals to enforce data submission

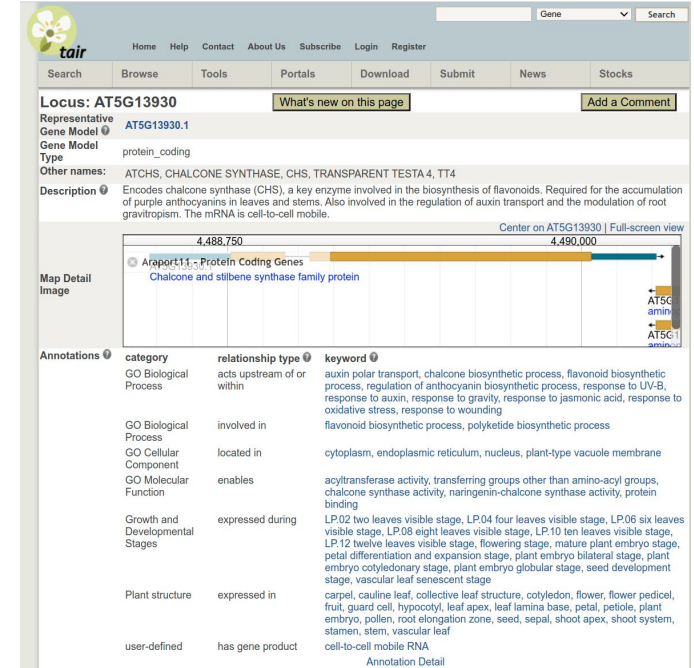


Species specific databases: TAIR

- TAIR is located at Phoenix Bioinformatics
- Institutional subscriptions provide funding for maintenance
- Full access requires subscription, but limited number of visits is free
- Collection of data sets and tools (e.g. BLAST)
- Current version: TAIR10 (=2010)
- Substitution by Araport11 was started, but is no longer funded



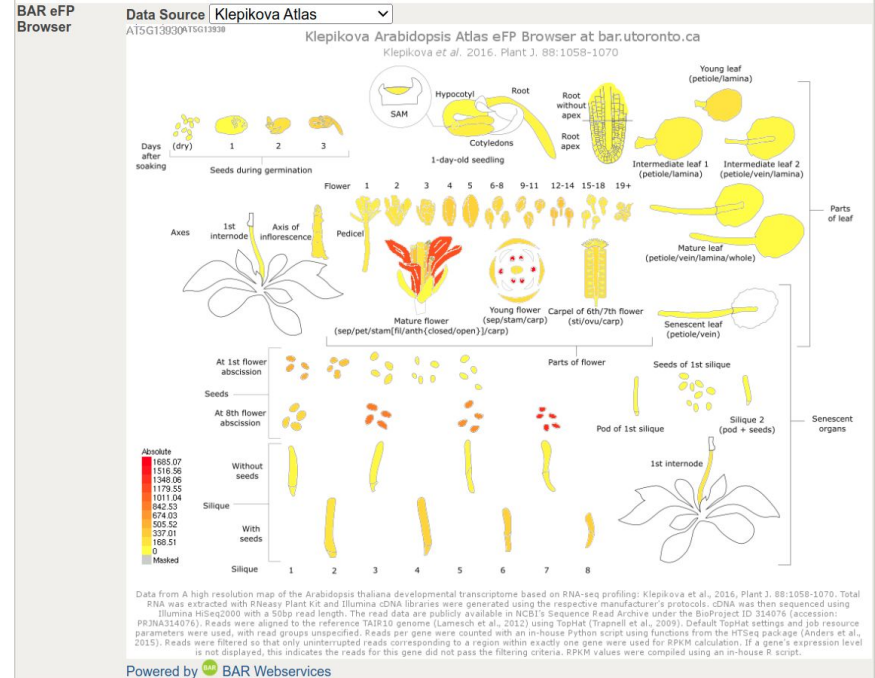
- TAIR10 provides information about function of genes
- References to other database for additional information
- Lists of publications about a given gene
- *Arabidopsis thaliana* gene functions are transferred to genes of other species



The screenshot displays the TAIR10 database interface for the gene AT5G13930. The page includes a navigation bar with links like Home, Help, Contact, and a search bar. The main content area shows the gene's locus, representative gene model (AT5G13930.1), and a description of its function as a chalcone synthase (CHS). Below the description is a map detail image showing the gene's location on a chromosome. The bottom section contains a table of annotations with columns for category, relationship type, and keyword.

category	relationship type	keyword
GO Biological Process	acts upstream of or within	auxin polar transport, chalcone biosynthetic process, flavonoid biosynthetic process, regulation of anthocyanin biosynthetic process, response to UV-B, response to auxin, response to gravity, response to jasmonic acid, response to oxidative stress, response to wounding
GO Biological Process	involved in	flavonoid biosynthetic process, polyketide biosynthetic process
GO Cellular Component	located in	cytoplasm, endoplasmic reticulum, nucleus, plant-type vacuole membrane
GO Molecular Function	enables	acyltransferase activity, transferring groups other than amino-acyl groups, chalcone synthase activity, naringenin-chalcone synthase activity, protein binding
Growth and Developmental Stages	expressed during	LP02 two leaves visible stage, LP04 four leaves visible stage, LP06 six leaves visible stage, LP08 eight leaves visible stage, LP10 ten leaves visible stage, LP12 twelve leaves visible stage, flowering stage, mature plant embryo stage, petal differentiation and expansion stage, plant embryo bilateral stage, plant embryo cotyledonary stage, plant embryo globular stage, seed development stage, vascular leaf senescent stage
Plant structure	expressed in	carpel, cauline leaf, collective leaf structure, cotyledon, flower, flower pedicel, fruit, guard cell, hypocotyl, leaf apex, leaf lamina base, petal, petiole, plant embryo, pollen, root elongation zone, seed, sepal, shoot apex, shoot system, stamen, stem, vascular leaf
user-defined	has gene product	cell-to-cell mobile RNA

- Gene expression analysis is integrated in TAIR
- eFP browser provides expression data
- eFP = electronic Fluorescent Pictograph
- Microarray and RNA-Seq data sets are basis of gene expression analysis

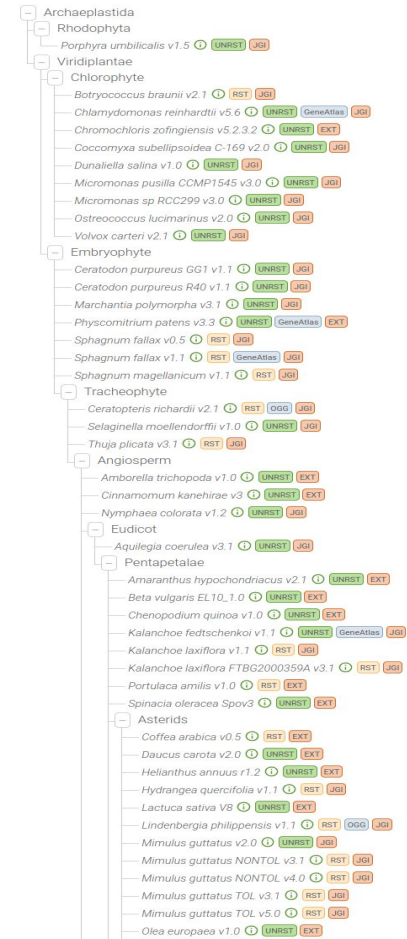


- Aggregates all references reporting about a gene
- Connection of knowledge
- AGI = Arabidopsis Gene Identifier
- AGIs mentioned in publication are used to assign studies to genes

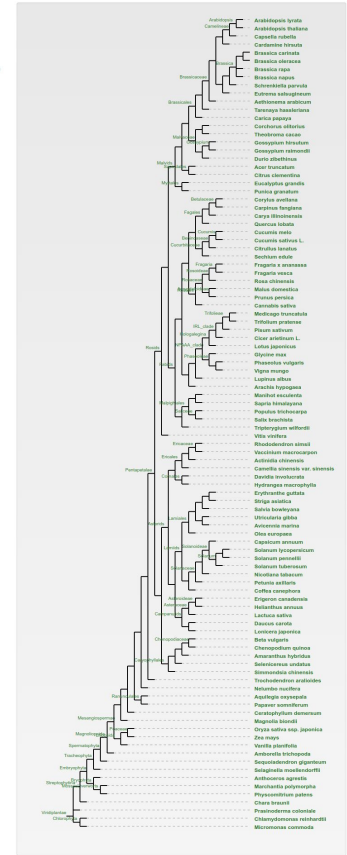
Publication #	author:title	source	associated loci	date
	Dahhan, D.A. Reynolds, G.D. Cair... Proteomic characterization of isolated Arabidopsis clathrin-coated vesicles reveals evolutionarily conserved and plant-specific components	THE PLANT CELL	AT2G46830 AT3G01780 AT3G51240 AT5G13930 AT5G61380	2022
	Verma, D. Bhagel, P.K. Sinha, A.K. A dual-specificity phosphatase, MAP kinase phosphatase 1, positively regulates blue light-mediated seedling development in Arabidopsis	PLANTA	AT1G29930 AT1G67090 AT2G04550 AT2G43790 AT2G46340 AT3G06110 AT3G23810 AT3G55270 AT4G36730 AT5G11250 AT5G13930 AT5G23720 AT5G24120 AT5G40440	2021
	Kissel, K.V. Suprun, A.R. Aleyno... External dsRNA Downregulates Anthocyanin Biosynthesis-Related Genes and Affects Anthocyanin Accumulation in Arabidopsis thaliana	INTERNATIONAL JOURNAL OF MOLECULAR SCIENCES	AT1G71030 AT1G77450 AT5G13930	2021
	Marhus, C. Pacinka, A. Merotto, A. Insights into the Role of Transcriptional Gene Silencing in Response to Herbicide-Treatments in Arabidopsis thaliana	INTERNATIONAL JOURNAL OF MOLECULAR SCIENCES	AT2G36490 AT5G13930	2021
	Zhang, X. He, Y. Li, L. Lu, H. Hon... Involvement of the R2R2-MYB transcription factors MYB21 and its homologs in regulating the stamen flavonoid accumulation in Arabidopsis	JOURNAL OF EXPERIMENTAL BOTANY	AT1G30135 AT2G47460 AT3G01530 AT3G27810 AT3G26210 AT5G08640 AT5G13930 AT5G40348 AT5G40350 AT5G46330 FL1	2021
	Yu, Z.C. Zheng, X.T. Lin, W. He... Photoprotection of Arabidopsis leaves under short-term high light treatment: The antioxidant capacity is more important than the anthocyanin shielding effect	PLANT PHYSIOLOGY AND BIOCHEMISTRY	AT1G177610 AT4G16880 AT4G22880 AT5G13930	2021
	Chapman, J.M. Muday, G.K. Flavonoids modulate lateral root emergence by scavenging reactive oxygen species in Arabidopsis thaliana	JOURNAL OF BIOLOGICAL CHEMISTRY	AT5G07990 AT5G08640 AT5G13930	2020
	Gayomba, S.R. Muday, G.K. Flavonoids regulate root hair development by modulating accumulation of reactive oxygen species in the root epidermis	DEVELOPMENT	AT5G13930 AT5G51060	2020
	Weng, C.Y. Zhu, M.H. Liu, Z.Q. Z... Integrated bioinformatics analyses identified SC1-induced regulatory network in Arabidopsis thaliana roots	BIOTECHNOLOGY LETTERS	AT1G50420 AT3G51240 AT3G55120 AT5G05270 AT5G13930	2020
	Naake, T. Maeda, H.A. Probst, S... Kingdom-wide analysis of the evolution of the plant type III polyketide synthase superfamily	PLANT PHYSIOLOGY	AT1G02050 AT4G34850 AT5G13930	2020
	Hammouch, N. Winkler, B.S.J. Li... Modulation of Arabidopsis Flavonol Biosynthesis Genes by Cyst and Root-Knot Nematodes	PLANTS (BASEL)	AT2G47460 AT5G08640 AT5G13930	2020
	Cheng, W. Lin, M. Qiu, M. Kong... Chitin synthase is involved in vegetative growth, asexual reproduction, and pathogenesis of Phytophthora capsici and P. sojae	ENVIRONMENTAL MICROBIOLOGY	AT3G33580 AT3G21630 AT5G13930	2019
	Nakayama, T. Takahashi, S. Waki, T. Formation of Flavonoid Metabolites: Functional Significance of Protein-Protein Interactions and Impact on Flavonoid Chemodiversity	FRONT PLANT SCI	AT5G13930	2019
	Rai, N. Neupart, S. Yan, Y. Wang... How do cryptochromes and UVR8 interact in natural and simulated sunlight?	JOURNAL OF EXPERIMENTAL BOTANY	AT5G13930 AT5G63860	2019

View Complete List (15 of 96 displayed)

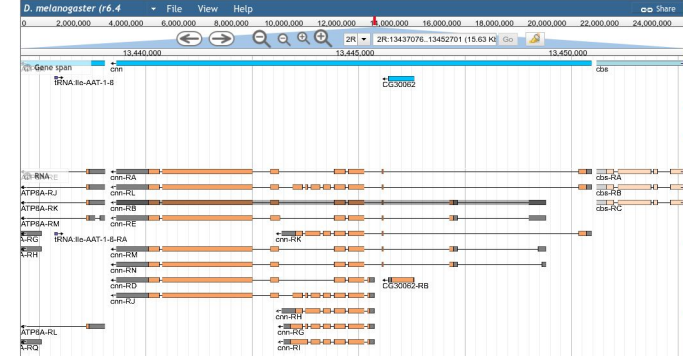
- Operated by the Joint Genome Institute (JGI), USA
- Collection of plant genome sequences and the corresponding annotations (best annotation source!)
- Data sets are available for download
- V13 contains 261 genome assemblies
- References of all data sets are clearly presented



- Hosted in Belgium (Department of Plant Biotechnology and Bioinformatics); partly funded by ELIXIR
- Collection of plant genome sequences and corresponding annotations
- Focus on comparative genomics
- PLAZA v5 contains 134 genome sequences and their annotations



- Website of the Drosophila research community
- BLAST service
- JBrowse = Web service to inspect the genome sequence
- Gene expression data; Data downloads
- Information about meetings, job opportunities, and courses



- Website of the *Cenorhabditis elegans* community
- Tools and resources are similar to FlyBase:
 - Option to run BLAST
 - Check of primers (e-PCR)
 - References to publications
 - Community news

e-PCR Search

This search uses Greg Schuler's [e-PCR](#) program to search for STSs in the current *C. elegans* genome.

You may also consider alternative tools to e-PCR, such as [UCSC In-Silico PCR](#) and [Primer-BLAST](#).

Enter a list of primer pairs to find using the following format:

Name	Left Oligo	Right Oligo	Length
assay_1	CGATAAACAATCAACGGCATAAT	TTTGAAACTGATATAGAGGGGCA	1188
assay_2	AAGGTTATTTATGCGGTGGAAAT	AGCACTTTGAGCTTGATGAAATC	2191
assay_3	AGATTGGAACGATAACGCAGATA	TTTGCCAATTTGCATTTATTTT	1683

Alternatively, select *List of assay names* and enter a list of PCR product names already in WormBase to see what genes they intersect.

e-PCR Search

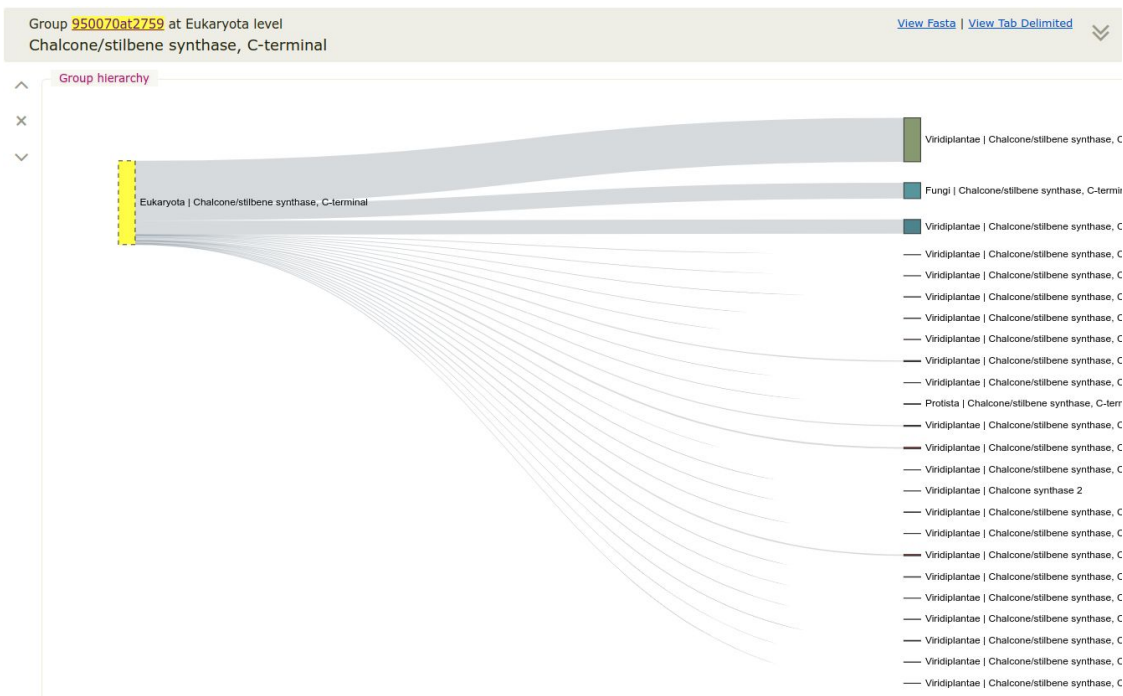
This is a ☒ List of primer pairs ☐ List of assay names

Allow product length difference of: bp Allow oligo mismatches of: bp

Results in: ☒ Text-only format ☐ HTML with hyperlinks

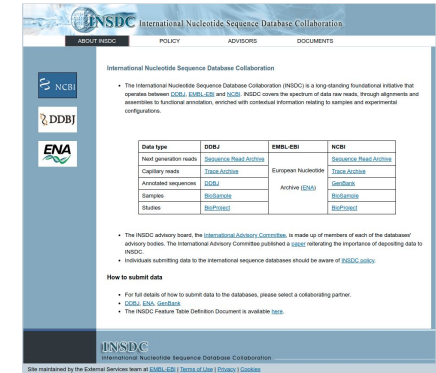
This may take several minutes to run. Hit the stop or back buttons to abort.

- Database of similar genes in other species



INSDC - a collaborative effort

- INSDC = International Nucleotide Sequence Database Collaboration
- Three databases are mirrored i.e. constantly updated
 - NCBI = National Center for Biotechnology Information (USA)
 - ENA = European Nucleotide Archive (Europe)
 - DDBJ = DNA DataBase of Japan (Japan)
- Users can search in any of the three databases and get the same results



- NCBI operates a large number of databases and services
- Famous service: BLAST
- Important databases:
 - PubMed: scientific publications
 - GenBank: individual sequences
 - Gene Expression Omnibus (GEO): gene expression data sets
 - Sequence Read Archive (SRA): sequencing data

Welcome to NCBI

The National Center for Biotechnology Information advances science and health by providing access to biomedical and genomic information.

[About the NCBI](#) | [Mission](#) | [Organization](#) | [NCBI News & Blog](#)

Submit

Deposit data or manuscripts into NCBI databases



Download

Transfer NCBI data to your computer



Learn

Find help documents, attend a class or watch a tutorial



Develop

Use NCBI APIs and code libraries to build applications



Analyze

Identify an NCBI tool for your data analysis task



Research

Explore NCBI research and collaborative projects



- BLAST = Basic Local Alignment Search Tool
- Probably the most famous website of the NCBI
- Comparison of sequences against a large database
- Numerous variants of the initial BLASTn were developed

The screenshot shows the NCBI BLAST homepage. At the top, it says 'Basic Local Alignment Search Tool' and provides a brief description of BLAST. Below this, there are links for 'Web BLAST' and 'BLAST Genomes'. The 'Web BLAST' section features three main tools: 'Nucleotide BLAST' (blastn), 'Protein BLAST' (blastp), and 'tblastn'. The 'BLAST Genomes' section has a search bar and a dropdown menu for selecting a genome (Human, Mouse, Rat, Microbes). Below this, there are links for 'Standalone and API BLAST' and 'Specialized searches'. The 'Specialized searches' section includes links to SmartBLAST, Primer-BLAST, Global Align, CD-search, IgBLAST, VecScreen, CDART, Multiple Alignment, and MOLE-BLAST, each with a brief description of its function.

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

BLAST+ 2.13.0 is here!
Starting with this release, we are including the blastn_vdb and tblastn_vdb executables in the BLAST+ distribution.
Thu, 17 Mar 2022 12:00:00 EST [More BLAST news...](#)

Web BLAST

Nucleotide BLAST
nucleotide → nucleotide

blastx
translated nucleotide → protein

tblastn
protein → translated nucleotide

Protein BLAST
protein → protein

BLAST Genomes

Enter organism common name, scientific name, or tax id

Human Mouse Rat Microbes [Search](#)

Standalone and API BLAST

[Download BLAST](#)
Get BLAST databases and executables

[Use BLAST API](#)
Call BLAST from your application

[Use BLAST in the cloud](#)
Start an instance at a cloud provider

Specialized searches

SmartBLAST
Find proteins highly similar to your query

Primer-BLAST
Design primers specific to your PCR template

Global Align
Compare two sequences across their entire span (Needleman-Wunsch)

CD-search
Find conserved domains in your sequence

IgBLAST
Search immunoglobulins and T cell receptor sequences

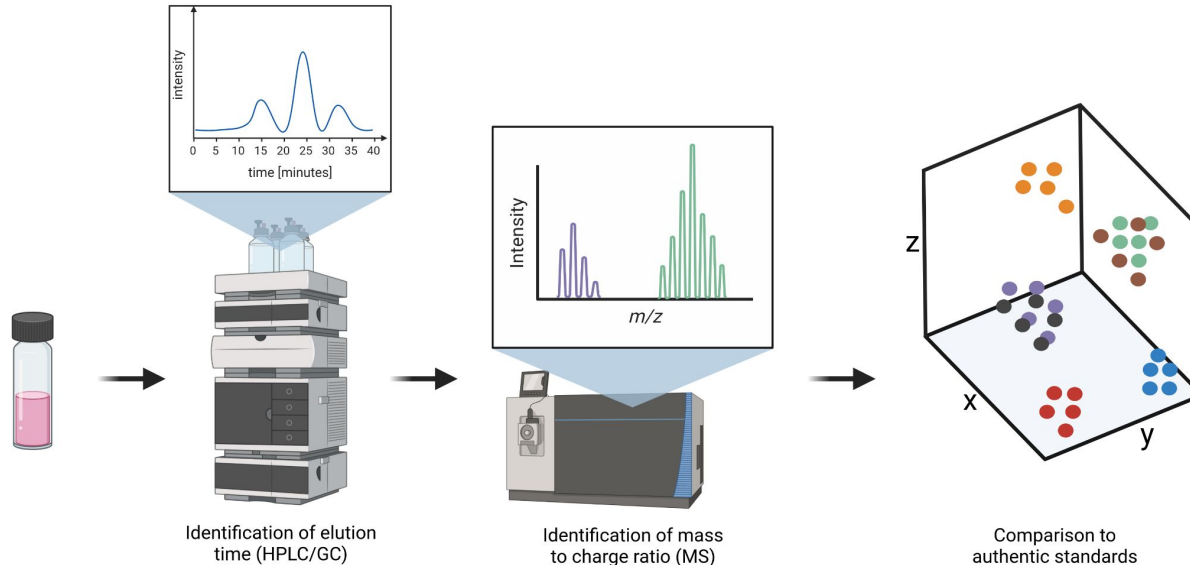
VecScreen
Search sequences for vector contamination

CDART
Find sequences with similar conserved domain architecture

Multiple Alignment
Align sequences using domain and protein constraints

MOLE-BLAST
Establish taxonomy for uncultured or environmental sequences

- Comparison of HPLC/GC-MS results
- Authentic standards required for identification of metabolite identity



- Search for studies based on metabolites or species
 - Example: find all studies associated with tyrosine
- Search for information about individual compounds
 - Delphinidin 3-O- β -D-glucoside
- Data sets are available for download

Download data from MetaboLights

 **MetaboLights ISAcreator software download.** To make it easy for new users, please download and just unzip our pre-packaged ISAcreator with plugin and configurations.

 **Experiments.** All public MetaboLights experiments can be downloaded from our public [ftp archive](#). Please find zip archives under the "studies" folder. Each public study can be found in the corresponding MTBLS-id folder. Complete experiments can be opened with [ISAcreator](#) or you can extract the archives using your normal unzip program.

Status: **Public** Release Date: **2016-08-23**
MTBLS333: Discovery of A-type procyanidin dimers in yellow raspberries by untargeted metabolomics and correlation based data analysis

Elisabete Carvalho, Pietro Franceschi

Introduction:
 Raspberries are becoming increasingly popular due to their reported health beneficial properties. Despite the presence of only trace amounts of anthocyanins yellow varieties seems to show similar or better effects in comparison to conventional raspberries.

Objectives:
 The aim of this work is to characterize the metabolic differences between red and yellow berries, focussing on the compounds showing a higher concentration in yellow varieties.

Methods:
 The metabolomic profile of 13 red and 12 yellow raspberries (of different varieties, locations and collection dates) was determined by UPLC-TOF-MS and a novel correlation based approach was implemented to extract the pseudospectra of the most relevant biomarkers from high energy LC-MS runs.

Results:
 Among the metabolites showing higher concentration in yellow raspberries it was possible to identify a series of compounds showing a pseudospectrum similar to that of A-type procyanidin polymers. The annotation of this group of compounds was confirmed by specific MS/MS experiments and performing standard injections.

Conclusion:
 In berries lacking anthocyanins the polyphenol metabolism might be shifted to the formation of a novel class of A-type procyanidin polymers.

MTBLC31463: Delphinidin 3-O-Beta-D-Glucoside

Chemistry **Biolog** **Spectra** **Literature**

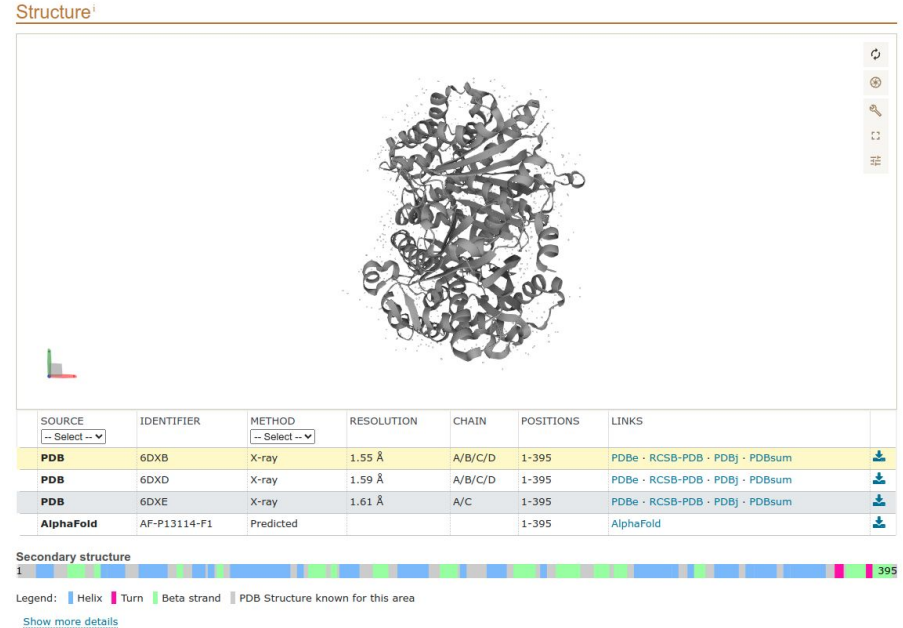
Compound Descriptors
 An anthocyan cation consisting of delphinidin having a beta-D-glucosyl residue attached at the 3-hydroxy position.

Identification

EUPAC Names	Delphinidin 3-O-glucoside
Molecular Formula	C ₂₇ H ₃₀ O ₁₂
Mass	466.18420
Monoisotopic Mass	466.103
Charge	+1
InChI	INCHI=1S[C@@H]1[C@H]([C@@H]([C@H]([C@H]([C@H]1O)O)O)O)O[C@H]2[C@H]([C@@H]([C@H]([C@H]([C@H]2O)O)O)O)O[C@H]3[C@H]([C@@H]([C@H]([C@H]([C@H]3O)O)O)O)O
InChIKey	KENPQOQDRAUJPEVUNPASA-O
SMILES	OC[C@H]1OC[C@H]([C@H]([C@H]([C@H]([C@H]1O)O)O)O)O[C@H]2[C@H]([C@@H]([C@H]([C@H]([C@H]2O)O)O)O)O[C@H]3[C@H]([C@@H]([C@H]([C@H]([C@H]3O)O)O)O)O
Synonyms	Delphinidin 3-O-Beta-D-Glucoside Delphinidin 3-Glucoside Delphinidin 3-O-Beta-D-Glucoside Delphinidin 3-O-Glucoside

Identification of enzymes - UniProt/SwissProt

- Curated database of protein sequences
- Provides additional information retrieved from other databases
- 2D and 3D structure of protein
- Sequence available for download

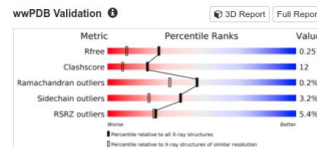


- Large collection of protein structures
- Download of protein structure files possible
- PDB viewer (DeepView) allows interactive inspection of protein structures

The screenshot shows the PDB website homepage. At the top, there's a search bar and navigation links. Below, the 'Advanced Search Query Builder' is visible, showing a search for 'chalcone synthase'. The 'Refinements' section on the left lists various filters like 'Release Date', 'Resolution', and 'R-Value'. The main content area displays a list of search results, with the first entry being '1EQY Chalcone isomerase and naringenin'.

Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 1.85 Å
R-Value Free: 0.262
R-Value Work: 0.237



This is version 1.2 of the entry. See complete history.

The screenshot shows the PDB viewer interface. On the left, there's a 'Sequence' tab showing the protein sequence. On the right, there's a 'Structure' tab showing a 3D ribbon diagram of the protein structure. The interface includes various controls for viewing and interacting with the structure.

- Chimera: <https://www.cgl.ucsf.edu/chimera/>
- PyMol: <https://pymol.org/2/>

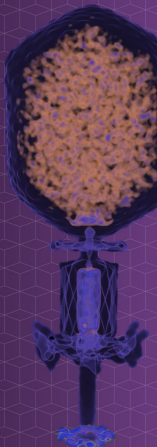
[illegible]

PyMOL is a user-sponsored molecular visualization system on an **open-source foundation**, maintained and distributed by **Schrodinger**.

We are happy to introduce
PyMOL 2.5!!

[DOWNLOAD NOW](#) [BUY LICENSE](#)

[RELEASE HIGHLIGHTS](#)



AlphaFold DB: Protein Structure Database

- AlphaFold is an Artificial Intelligence tool developed by DeepMind
 - Leading tool in the prediction of protein structures
- Corresponding database is maintained by EMBL-EBI
- Database contains predicted protein structures

Chalcone synthase

AlphaFold structure prediction

Download [PDB file](#) [mmCIF file](#) [Predicted aligned error](#)
[Feedback on structure](#) [Looks great](#) [Could be improved](#)

Information

Protein Chalcone synthase
 Gene Unknown
 Source organism Hypericum androsaemum (Tutani) [go to search of](#)
 UniProt Q9FUB7 [go to UniProt of](#)
 Experimental structures None available in the PDB
 Biological function The primary product of this enzyme is 4,2',4''-trihydrochalcone (also termed naringenin-chalcone or chalcone) which undergoes enzyme-catalyzed or spontaneous isomerization into naringenin. [go to UniProt of](#)

3D viewer

Model Confidence:

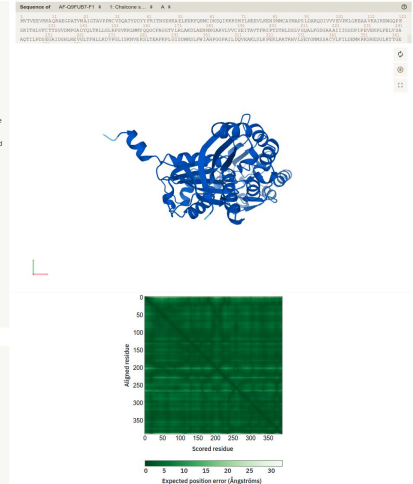
- Very high (pLDDT > 90)
- Confident (90 > pLDDT > 70)
- Low (70 > pLDDT > 50)
- Very low (pLDDT < 50)

AlphaFold produces a per-residue confidence score (pLDDT) between 0 and 100. Some regions below 50 pLDDT may be unstructured in isolation.

Views

[Predicted aligned error](#)

Sequence Features (coming soon)



Predicted aligned error

The colour at position (x, y) indicates AlphaFold's expected position error at residue x, when the predicted and true structures are aligned on residue y.

This is useful for assessing inter-domain accuracy - see the tutorial below.

[Predicted aligned error tutorial](#)

Identification of enzymes - InterProScan5

- Screen of protein sequences against collection of protein signatures
- Allows the assignment of functional annotation terms
- Available as web service, but also as stand alone tool

InterProScan 5 Sequence Search

This form allows you to scan your sequence for matches against the InterPro collection of protein signature databases.

This form is for debugging purposes only and is **not supported**. To submit jobs to InterProScan 5, please visit the [InterPro Sequence Search](#) or the [InterProScan 5 Web services](#).

Please Note

This web form is for internal debugging purposes and will be retired in April 2016. If you have any questions/concerns please contact us via the feedback link above.

STEP 1: Enter your input sequence

Enter or paste a **PROTEIN** sequence in any supported format:

uniprot:KPYM_HUMAN

Or, upload a file (Choose file) No file chosen

Use an example sequence | Clear sequence | See more example inputs

STEP 2: Select the applications to run

☒ Pfam ☒ PIRSF ☒ Probus ☒ SignalP ☒ SignalP_EUK
☒ SignalP_GRAM_POSITIVE ☒ SignalP_GRAM_NEGATIVE ☒ SUPERFAMILY ☒ PANTHER ☒ Gene3D
☒ Pfam ☒ Probus ☒ ProSitePatterns ☒ CDD ☒ PROSITE ☒ PROSITE
☒ TIGRfam ☒ PROSITE ☒ Pfam ☒ MAADBLite ☒ PROSITE

STEP 3: Submit your job

☐ Be notified by email (Tick this box if you want to be notified by email when the results are available)

Submit

Results for job iprscan5-I20220320-102557-0980-87120812-p2m

Tool Output

Submission Details

Download in XML format

Download in TSV format

Download in GFF3 format

Download in SVG format

Download HTML tarball file

Download in JSON format

sp P13114 CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	Gene3D	G3DSA:3.40.47.10	-	1	241	4.8E-101	T	20-03-2022
sp P13114 CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	SUPERFAMILY	SSF53901	Thiolase-like	241	393	2.98E-51	T	20-03-20
sp P13114 CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	SUPERFAMILY	SSF53901	Thiolase-like	10	237	2.38E-78	T	20-03-20
sp P13114 CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	Pfam	PF02797	Chalcone and stilbene synthases, C-terminal domain	244	394	1.5E-71	T	20-03-20
sp P13114 CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	PANTHER	PTHR11877:SF81	BNA02G30320D PROTEIN	6	394	0.0	T	20-03-2022
sp P13114 CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	Gene3D	G3DSA:3.40.47.10	-	242	395	9.7E-62	T	20-03-2022
sp P13114 CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	PANTHER	PTHR11877	HYDROXYMETHYLGLUTARYL-COA SYNTHASE	6	394	0.0	T	20-03-2022
sp P13114 CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	CDD	cd00831	CHS_like	21	390	0.0	T	20-03-2022
sp P13114 CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	PIRSF	PIRSF000451	PKS III	7	394	0.0	T	20-03-2022
sp P13114 CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	Pfam	PF00195	Chalcone and stilbene synthases, N-terminal domain	10	233	2.9E-119	T	20-03-2022
sp P13114 CHSY_ARATH	ba46b0f06bfbf1c91d161f0f91310f43	395	ProSitePatterns	PS00441	Chalcone and stilbene synthases active site.	161	177	-	T	20-03-2022

- Defined statements about the function of a gene (controlled vocabulary)
- Hierarchical structure
 - Example: 'metabolic process' > 'biosynthetic process' > ... > 'chalcone synthase'
- Supported by the Alliance of Genome Resources
- Connected with various other databases e.g. TAIR, FlyBase, Reactome, UniProt
- Machine readable to allow automatic processing
- Tools: Blast2GO and AmiGO
 - analyze the function of a sequence (web service and standalone)

```

1 GO:0008150 biological_process
1 GO:0008152 metabolic process
1 GO:0009058 biosynthetic process
1 GO:0071704 organic substance metabolic process
1 GO:0009812 flavonoid metabolic process
1 GO:1901576 organic substance biosynthetic process
  ▼ GO:0009813 flavonoid biosynthetic process
    1 GO:0009718 anthocyanin-containing compound biosynthetic process
    1 GO:0051551 aurone biosynthetic process
    1 GO:0033485 cyanidin 3-O-glucoside biosynthetic process
    1 GO:0033486 delphinidin 3-O-glucoside biosynthetic process
    1 GO:0051553 flavone biosynthetic process
    1 GO:0009716 flavonoid phytoalexin biosynthetic process
    1 GO:0051557 leucoanthocyanidin biosynthetic process
    1 GO:0009964 negative regulation of flavonoid biosynthetic process
    1 GO:0033487 pelargonidin 3-O-glucoside biosynthetic process
    1 GO:0009963 positive regulation of flavonoid biosynthetic process
    1 GO:0009962 regulation of flavonoid biosynthetic process
  
```


- AmiGO allows text or sequence searches
- Options to explore annotations in detail
- Database covers various kingdoms (not restricted to plants)

Total term(s): 6; showing: 1-6
Results count: 10

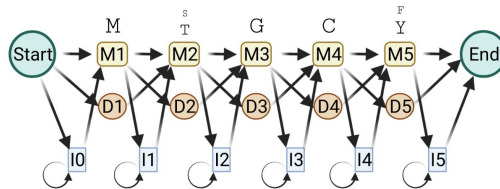
«First» «Prev» «Next» «Last» Custom DL (up to 100000) Bookmark

☐ Term	Definition	Ontology source	Ontology ID space	Synonyms	Alt ID
☐ flavonoid glucuronidation	The modification of a flavonoid by the conjugation of glucuronic acid. The resultant flavonoid glucu more...	biological_process	GO	flavonoid glucuronide biosynthesis flavonoid glucuronide biosynthetic process more...	
☐ flavonoid phytoalexin biosynthetic process	The chemical reactions and pathways resulting in the formation of flavonoid phytoalexins, a group of more...	biological_process	GO	flavonoid phytoalexin anabolism flavonoid phytoalexin biosynthesis flavonoid phytoalexin formation flavonoid phytoalexin synthesis	
☐ flavonoid biosynthetic process	The chemical reactions and pathways resulting in the formation of flavonoids, a group of phenolic derivatives containing a flavan skeleton.	biological_process	GO	flavonoid anabolism flavonoid biosynthesis flavonoid formation flavonoid synthesis	
☐ regulation of flavonoid biosynthetic process	Any process that modulates the frequency, rate or extent of the chemical reactions and pathways resulting in the formation of flavonoids.	biological_process	GO	regulation of flavonoid anabolism regulation of flavonoid biosynthesis more...	
☐ negative regulation of flavonoid biosynthetic process	Any process that stops, prevents, or reduces the frequency, rate or extent of the chemical reactions more...	biological_process	GO	down regulation of flavonoid biosynthetic process down-regulation of flavonoid biosynthetic process more...	
☐ positive regulation of flavonoid biosynthetic process	Any process that activates or increases the frequency, rate or extent of the chemical reactions and pathways resulting in the formation of flavonoids.	biological_process	GO	positive regulation of flavonoid anabolism positive regulation of flavonoid biosynthesis more...	

- Assignment of protein functions based on Hidden Markov Models (HMMs)
- Sequences are screened based on HMM profile

Sequences are aligned
↓
Modelling of insertions/deletions
↓
Amino acid frequency at each alignment position is encoded
↓
HMM profile generated

seq1	MTGC-Y	2 = deletion
seq2	MSGC-F	5 = insertion
seq3	MTGC-Y	
seq4	M-GCAY	
	123456	



[HOME](#) | [SEARCH](#) | [BROWSE](#) | [FTP](#) | [HELP](#) | [ABOUT](#)

Pfam 35.0 (November 2021, 19632 entries)

The Pfam database is a large collection of protein families, each represented by **multiple sequence alignments** and **hidden Markov models (HMMs)**. [More...](#)

QUICK LINKS	YOU CAN FIND DATA IN PFAM IN VARIOUS WAYS...
SEQUENCE SEARCH	Analyze your protein sequence for Pfam matches
VIEW A PFAM ENTRY	View Pfam annotation and alignments
VIEW A CLAN	See groups of related entries
VIEW A SEQUENCE	Look at the domain organisation of a protein sequence
VIEW A STRUCTURE	Find the domains on a PDB structure
KEYWORD SEARCH	Query Pfam by keywords
JUMP TO	Enter any accession or ID Go Example
	Enter any type of accession or ID to jump to the page for a Pfam entry or clan, UniProt sequence, PDB structure, etc.
	Or view the help pages for more information

Recent Pfam [blog](#) posts

[Pfam 35.0 is released](#) (posted 19 November 2021)

Pfam 35.0 contains a total of 19,632 families and clans. Since the last release, we have built 460 new families, killed 7 families and created 12 new clans. UniProt Reference Proteomes has increased by 7% since Pfam 34.0, and now contains 61 million sequences. Of the sequences that are in UniProt Reference Proteomes, 75.2% have [...]

[AlphaFolding the Protein Universe](#) (posted 22 July 2021)

Hot on the tail of our inclusion of the Baker group's trRosetta structural models we are excited to announce the inclusion of models from AlphaFold 2.0 generated by DeepMind and stored in the AlphaFold Database (AlphaFold DB). AlphaFold 2.0's performance in the CASP14 competition was spectacular, producing near experimental quality structure models. The new AlphaFold [...]

[Google Research Team bring Deep Learning to Pfam](#) (posted 24 March 2021)

We are delighted to announce the first fruits of a collaboration between the Pfam team and a Google Research team led by Dr Lucy Colwell, with Maxwell Bileschi and David Belanger. In 2019, Colwell's team published a preprint describing a new deep learning method that was trained on Pfam data, and which improves upon the [...]

Citing Pfam

If you find Pfam useful, please consider [citing](#) the reference that describes this work:

Pfam: The protein families database in 2021 (Mistry, S., Chuguransky, L., Williams, M., Qureshi, G.A., Salazar, E.L.L., Sonnhammer, S.C.E., Tosatto, L., Paladini, S., Raj, L.J., Richardson, R.D., Finn, A., Bateman)

Nucleic Acids Research (2020) doi: 10.1093/nar/gkaa913

- Online tool for the annotation of all protein sequences in a submitted FASTA file
- FASTA format:
 - Header line starts with '>' followed by the sequence name
 - Header is followed by unrestricted number of sequence lines

```
>TRINITY_DN100013_c0_g1_i1
MIGPMPGMEGKMLPGASPVGLEVVLSLVVASTPILSDTSLCTPSFYHFLLLLGPITSISNLIIVRPFFLTLLSSITVIYGRLCFFAFDFYVY
>TRINITY_DN100016_c0_g1_i1
MPKKSSNIQQNNQNRIKRKGRPPKKKKYVQQIDSSDEDILSVRHASTRPRIISIRRNEIPMRPEIHIPPTSIPSFNNPNNIHQSSSDRQMPP
>TRINITY_DN100019_c0_g1_i1
MPPKAADKKPAAPATASKAPEKKDAGKKAATGDKKKRTKARKETYSSYIYKVLKQVHPDTGISNRAMSILNSFVNDIFERVATEASKLA
>TRINITY_DN100021_c0_g1_i2
MAKVGNPIVIDIEDTGSVNEPESSEKNIEVSSSSTQAPESTNTTELLVNEKKAFSLATPAVRRVAREHNIDINNIKGTGKNGRITKEDILNYV
>TRINITY_DN100025_c0_g1_i1
MVENQDGCFFKPGWKEFVRSNDLEGGDFLVFNLVDKISYQVVIFDGTCCPKDLCPSIMNPFIQHLRNKIFLSKKEEIKLKGNRKVHSVNE
>TRINITY_DN10002_c0_g1_i1
MSDELNQVFQRCREQKRPVFVAFVTAGYPDSEETVDILLGLEAGGADIIELGIPFTDPMVDGKTIQDANNVALENKIDISKCLSYVSESRAK
```

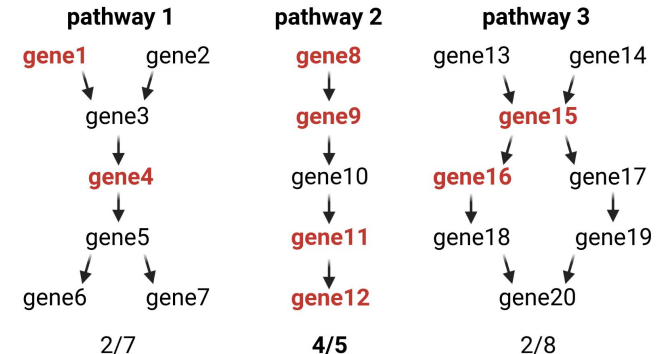
Concept of enrichment analyses

- Omics analyses can identify a large set of differentially expressed genes
- Making sense of large gene sets is challenging
- Identification of a common function is necessary to reduce complexity
- Enrichment analyses can help to identify shared functions or pathways

Differentially expressed genes (DEGs)

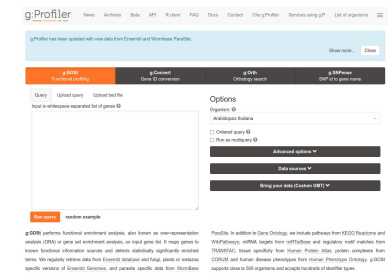
gene1
gene4
gene8
gene9
gene11
gene12
gene15
gene16
gene22
...

↓ Identification of enriched pathways

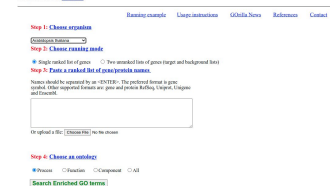


Enrichment analyses - GO

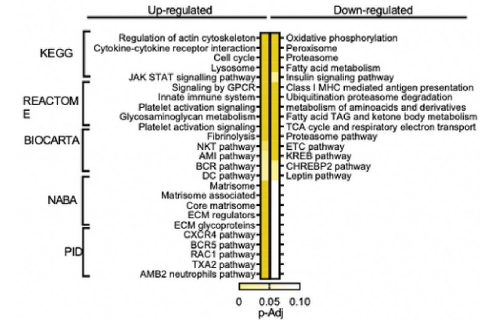
- Functional enrichment analyses can be based on GO terms
- Assignment of GO terms to genes of the species of interest
- Tools for GO enrichment analysis:
 - PANTHER: <http://geneontology.org/>
 - g:Profiler: <http://biit.cs.ut.ee/gprofiler/gost>
 - GOrilla: <http://cbl-gorilla.cs.technion.ac.il/>



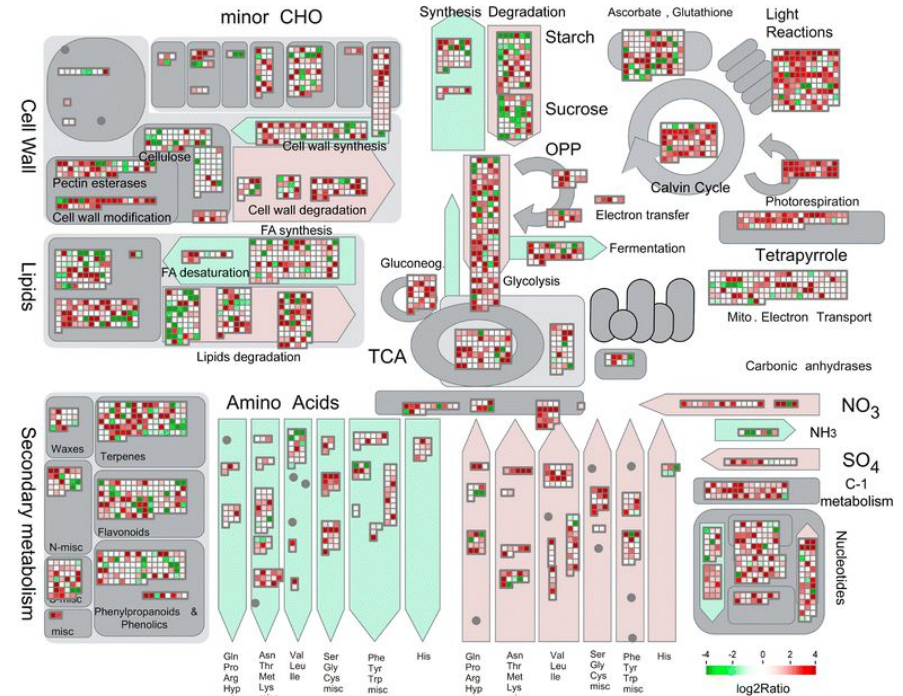
GOrilla is a tool for identifying and visualizing enriched GO terms in ranked lists of genes. It can be used for: 1. Identifying the enriched GO terms in a ranked list of genes. 2. Visualizing the enriched GO terms in a ranked list of genes.



- Identification of enriched KEGG pathways among a gene set
- Available tools:
 - KEGG Mapper: <https://www.genome.jp/kegg/kegg1b.html>
 - pathfinder: <https://doi.org/10.3389/fgene.2019.00858>
- Names of pathways is not always informative
- KEGG pathways can be too superficial and are only a first step in analyses



- Identification of enriched pathways
- Visualization of up- and down-regulated pathways:
 - Red = up-regulation
 - Green = down-regulation
 - White = neutral (no change)
- Overview of the metabolism of a cell
- MapMan:
<https://mapman.gabipd.org/mapman>



- Types of databases (reactions, enzymes, sequences, metabolites, species-specific)
- Species-specific databases
- Finding information about enzymes, transcription factors, and other genes
- Finding information about protein structures
- NCBI & INSDC
- Enrichment analyses

Time for questions!

1. What is the meaning of FAIR data?
2. How would you get insights into the function of an unknown sequence?
3. How can you identify the enzymes involved in the flavonol biosynthesis?
4. Where can you find information about the substrate affinity of the chalcone synthase?
5. What is TAIR?
6. Where can you find information about the expression of *MYB12* in different plant organs?
7. Where can you get the genome sequence and corresponding annotation of *Marchantia polymorpha*?
8. Which databases are connected through INSDC?
9. Where can you find the results of metabolomic studies?
10. Where can you find information about the structure of CHS?
11. Which tools can be applied to annotate large sets of (poly)peptide sequences?
12. Which tools can be applied for pathway enrichment analyses?
13. What are the two line types in a FASTA file?